



Mohamed Dhia BEN HMIDA

Software Development Engineer

Contact

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Summary

Software engineer with 7 years of experience in designing critical systems for the financial sector. Specialized in Java/Spring Boot and Oracle SQL/PL-SQL (packages, jobs, materialized views, partitioning, indexing, execution plan tuning), with proficiency in Python/FastAPI and React. Accustomed to high-volume environments with strict compliance requirements, focusing on performance, data quality, security (RBAC, encryption in transit/at rest), and observability.

Focused on regulatory and accounting solutions for financial institutions: data collection and normalization architecture, integrity and traceability controls, batch automation and ETL pipelines, prudential reporting generation, and industrialization via CI/CD and containerization. Proven experience in performance optimization (latency, volume), data governance, and deploying auditable deliverables.

Skills

Programming:

Python, JavaScript, TypeScript, PL/SQL, Java, PHP, VBA

Professional Experience

Java and Oracle PL/SQL Developer

BFI Groupe

July 2023 – Present

- Designed and developed robust software solutions for financial institutions, aligning with industry standards and specifications.
- Accounting centralizer for BIAT: integration and synchronization with the bank's information systems (core banking and other flows) via the accounting centralizer; orchestration of accounting and transactional flows; integrity and traceability controls; performance optimization (latency, volume).
- Regulatory application for banks in the UEMOA region: design and integration of a prudential reporting solution compliant with BCEAO/UEMOA requirements (collection, normalization, controls, generation, and transmission of reports).
- Collaborated with cross-functional teams to analyze requirements, architect scalable solutions, and solve complex technical problems throughout the development lifecycle.
- Implemented and optimized high-performance Oracle databases, focusing on data integrity and security to support critical transactions.
- Developed custom features, including reports, automated workflows, and user access controls tailored to each client's unique needs.
- Wrote comprehensive technical documentation, including architecture diagrams, API references, and user manuals.

Development Engineer

SEGULA Technologies for Stellantis Group

October 2018 – June 2023

- Led the analysis and detailed specification of client requirements to ensure the delivery of high-impact solutions.
- Architected and developed advanced internal software for the objective calibration of automatic transmission torque converters, leveraging vehicle dynamics with Excel VBA as a foundation.
- Designed and implemented a comprehensive benchmarking tool that automatically generates detailed .doc reports based on complex client-defined criteria.

Web Development:

React, FastAPI, Express (Node.js), Spring Boot, HTML, Bootstrap, Material UI

State Management:

Redux Toolkit

Databases:

PostgreSQL, MongoDB, MySQL, Oracle SQL, JPA/Hibernate, Mongoose, Redis, IndexedDB, LocalStorage

Machine Learning & AI:

PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Vision Transformers (ViT), EfficientNet, ConvNext, TIMM, LLMs, MediaPipe, OpenCV, TensorBoard

3D Graphics & CAD:

Three.js, React Three Fiber, OpenCASCADE.js, WebAssembly (WASM), Parametric Modeling, CAD File Formats (STEP, IGES, STL, OBJ)

Methodologies:

Agile Scrum

CI/CD & Tools:

Git, Docker, Vite, WebSocket, WebRTC, JWT Authentication

Languages

Arabic: Native

French: Fluent

English: Fluent

Education

National Diploma of Engineering

National School of Engineers of Tunis (ENSIT)

September 2015 – June 2018

Preparatory Classes for Engineering Studies (MP track)

Preparatory Institute for Engineering Studies of El Manar (IPEIEM), Tunis

2013 – 2015

Tunisian Baccalaureate — Mathematics specialty

2013

- Engineered a robust Excel/Access platform for data navigation, visualization, and updating, with external XML import/export and seamless synchronization with an Access database via optimized SQL queries; later migrated to PostgreSQL while ensuring backward compatibility with existing tools.

Featured Projects

SkinAI

- Developed a full-stack AI-based skin analysis application using React (frontend), Python FastAPI and Spring Boot (backend), and PostgreSQL.
- Implemented secure user authentication, analysis history, and AI image analysis using custom machine learning models.

CADSHIP

- Developed a full-stack web platform for managing and visualizing CAD files (STL, OBJ).
- Node.js/Express backend for file processing and a React frontend for an interactive user interface.

Centralized AI Model Framework

- Architecture: React (UI), FastAPI/Spring Boot API, model registry with a unified inference interface (adapters), PostgreSQL, Docker.
- Features: seamless model swapping, interchangeability in workflows/pipelines without modifying consumer code, versioning and rollback, A/B testing, and change auditing.

Training & Certifications

Accelerated VBA Training - Stellantis

Technical Center, Carrières-sous-Poissy

August 2019

- Intensive practical training on advanced VBA development in industrial applications.

Massinissa Framework Training - BFI

Proprietary Java MVC Framework

4 months

- 4-month program on the architecture of the Massinissa framework, based on Java and the MVC architecture.